this. “But you don’t seriously believe that none but observable magnitudes must go into a physical theory?”

“Isn’t that precisely what you have done with relativity?” replied Heisenberg.

“Possibly I did use this kind of reasoning, but it is nonsense all the same,” said Einstein. “On principle, it is quite wrong to try founding a theory on observable magnitudes alone. In reality the very opposite happens. It is the theory which decides what we can observe.” Einstein then went on to explain that the information about the world around us that we receive from scientific instruments—or even from our own senses—would be totally incomprehensible without some kind of theory about the way the world works. When you use a thermometer to test the temperature of a chicken you’ve cooked in the oven, you’re assuming the thermometer accurately indicates the temperature inside of the chicken—and that the light that reflected off of the thermometer and entered your eyes accurately indicates the reading of the thermometer. In other words, you have a theory about how the world works, and you’re using that (very well-justified!) theory to inform your use of the thermometer. Similarly, Einstein pointed out to Heisenberg that, when looking at the spectrum of an atom, “you quite obviously assume that the whole mechanism of light transmission from the vibrating atom to the spectroscope or to the eye works just as one has always supposed it does.”

Heisenberg was “completely taken aback by Einstein’s attitude,” as he recalled later. Falling back on the seemingly solid ground of Mach’s philosophy, Heisenberg replied, “The idea that a good theory is no more than a condensation of observations… surely goes back to Mach, and it has, in fact, been said that your relativity theory makes decisive use of Machian concepts. But what you have just told me seems to indicate the very opposite. What am I to make of all this, or rather what do you yourself think about it?”

“Mach rather neglects the fact that the world really exists, that our sense impressions are based on something objective,” Einstein replied. “He pretends that we know perfectly well what the word ‘observe’ means, and that this exempts him from having to discriminate between ‘objective’ and ‘subjective’ phenomena.… I have a strong suspicion that, precisely because of the problems we have just been discussing, your theory will one day get